

Psychiatric disorders

Anxiety, Depression and Psychosis

Integrated Biomedical and Clinical Sciences OMFS 512

Shymaa Bilasy, pharm B, Ph.D.

Mironda Frankenberger, DDS



Mission: To advance the science and art of health through education, service, scholarship and social accountability
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1

Learning Objectives

Learning Outcomes

1. Describe the prevalence of mental illness.
2. Describe physiological and psychological aspects of depression.
3. Describe physiological and psychological aspects of anxiety .
4. Describe physiological and psychological aspects of schizophrenia.

Outline

1. Prevalence of mental illness
2. Mood disorders
3. Anxiety disorders
4. Psychotic Disorders (Schizophrenia)

Prevalence of Mental Illness

23.4%

of adults in the U.S. **experienced any mental illness** (AMI) in the past year, equivalent to over 60 million people.

17.7%

of adults in the U.S. **had a substance use disorder** (SUD) in the past year, totaling over 46 million people.

5.5%

of adults **reported experiencing serious thoughts of suicide**, totaling over 14 million people.

Mental health among youth in the U.S. improved significantly from 2023 to 2024.



The percentage of **youth** (ages 12-17) who experienced a **major depressive episode** (MDE) in the past year **decreased** from 18.10% in 2023 to 15.40% in 2024.



The percentage of **youth** (ages 12-17) reporting **serious thoughts of suicide** in the past year **decreased** from 12.30% of youth in 2023 to 10.10% of youth in 2024.

<https://mhanational.org/the-state-of-mental-health-in-america/>

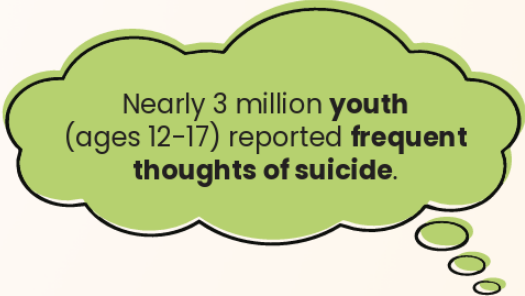
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6

Prevalence of Mental Illness

But still, young people are struggling.
In 2024:

11.3% of **youth** (ages 12-17) in the U.S. experienced a **major depressive episode (MDE) with severe impairment in the past year**, meaning it severely impacted their functioning at work, school, or home. That's a total of 2.8 million youth whose depression significantly impaired their functioning.



Nearly 3 million **youth** (ages 12-17) reported **frequent thoughts of suicide**.

<https://mhanational.org/the-state-of-mental-health-in-america/>

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7

Mental illnesses include different conditions with varying severity.

Any mental illness (AMI)

Broad umbrella that encompasses all recognized mental illnesses. Mental illness encompasses mental, behavioral, or emotional disorders.

AMI impact: Variable, from no impairment-severe impairment.

Among patients with mental illnesses, 4.91% are experiencing a serious mental illness.

**1 in each 5 adults
has “any mental
illness”**

Serious mental illness (SMI)

Mental, behavioral, or emotional disorder resulting in serious functional impairment that substantially interferes with or limits one or more major life activities.

For example, schizophrenia, bipolar depression, major depression.

**1 in each 20 adults
has “serious
mental illness”**

Categories of Mental Illness

Neurocognitive
disorders

Mood
disorders

Anxiety
disorders

Psychosis

Dissociative
disorders

Posttraumatic
stress disorder

Obsessive-
compulsive
disorder

Eating
disorders

Sleep disorders

Impulse-
control
disorders

Developmental
disorders

Substance-
related
disorders

Personality
disorders

Child
Maltreatment

Autism and
Autism
Spectrum
disorders

Role of Neurotransmitters and Genetics

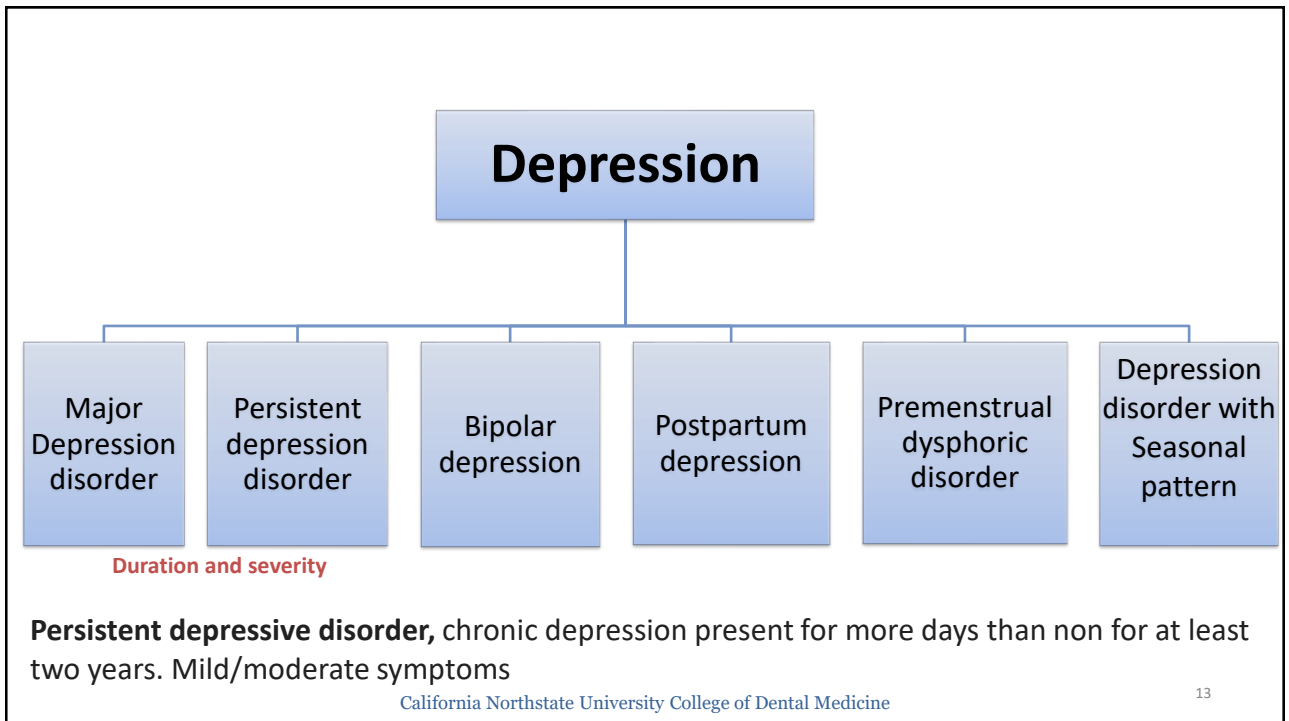
- Specific gene variants were identified to increase the risk for schizophrenia, bipolar disorder, alcohol use disorder, intellectual disability and others.
- Clinical studies revealed alterations in neurotransmitter levels in patients with certain mental illnesses.

Mood Disorders

Depression and Bipolar Depression

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12

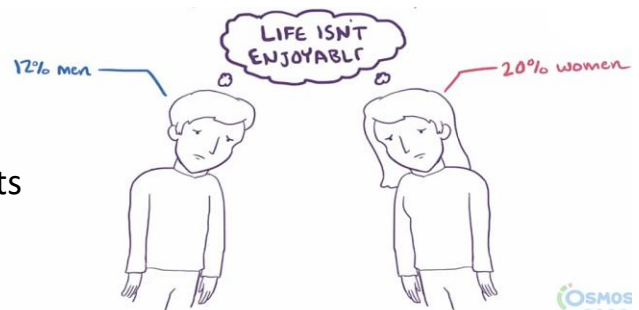


Major Depression disorder

Unipolar depression

Major depression (aka, clinical depression or unipolar depression) is a common mental health illness worldwide (280 million persons).

In USA, around 21.0 million adults had at least one major depressive episode (8.4% of U.S. adults).
Higher prevalence in females.



OSMOS
2022

Causes of Major Depression Disorder (MDD)

Genetic, biological, environmental, and psychological factors play a role in depression.

- **Personal or family history of depression.**

Heritability around 35–40%.

Gene deficiencies in serotonin transporter gene (*5-HTT*) and serotonin receptor gene *HTR2A* can increase the risk of depression.

- **Stress, addiction, sedentary lifestyle**

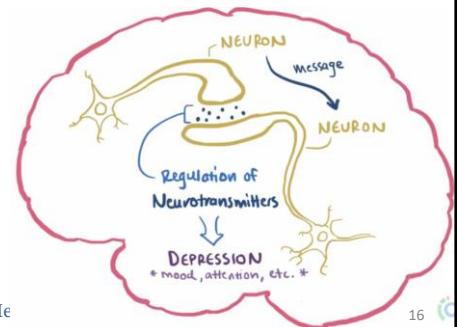
- **Certain medical conditions** (diabetes, cancer, cardiovascular disease, addiction).

- **Substance/medications use** (illicit substances, alcohol, hormones).

Pathophysiology of MDD

Multiple theories.

- ❖ Impact of altered brain structure and function. For example, cerebellar damage or abnormal structure can be linked to depression.
- ❖ Brain neurotransmitters can impact different brain functions.
 - For example, depression symptoms were improved by increasing the synaptic concentrations of monoamine neurotransmitters. This led to the development of the monoamine hypothesis.



Diagnosis of MDD

- According to “The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)”, five or more depression symptoms are required for the diagnosis of depression.
- Symptoms must be present most of the day, nearly every day, for 2 weeks.
- Symptoms cause clinically significant distress or impairment in functionality (like, social, occupational).
- The episode is not attributed to the physiologic effects of a substance or to another medical condition.
- The absence of a manic episode or a hypomanic episode.

For MDD diagnosis, symptoms must include depressed mood and/or anhedonia

Symptoms

- Depressed mood (feeling sad, irritable, empty).
- Loss of pleasure or interest in activities (anhedonia) .
- Sleep disorder (insomnia or hypersomnia).
- Changes in appetite or weight.
- Fatigue
- Trouble concentrating.
- Low self-esteem or guilt.
- Psychomotor retardation or agitation.
- Recurrent thoughts of death or suicidal thoughts.

Five or more
symptoms must be
present most of the
day, nearly every
day, for 2 weeks

Management strategies for MDD

- Pharmacological approaches

Medications that modulate neurotransmitter levels.

- Non-pharmacological approaches

Diet, lifestyle and psychotherapy (like cognitive behavioral therapy, CBT).

For refractory depression: Electroconvulsive therapy (ECT) can be an alternative therapeutic option.

Bipolar depression

manic depression or bipolar disorder

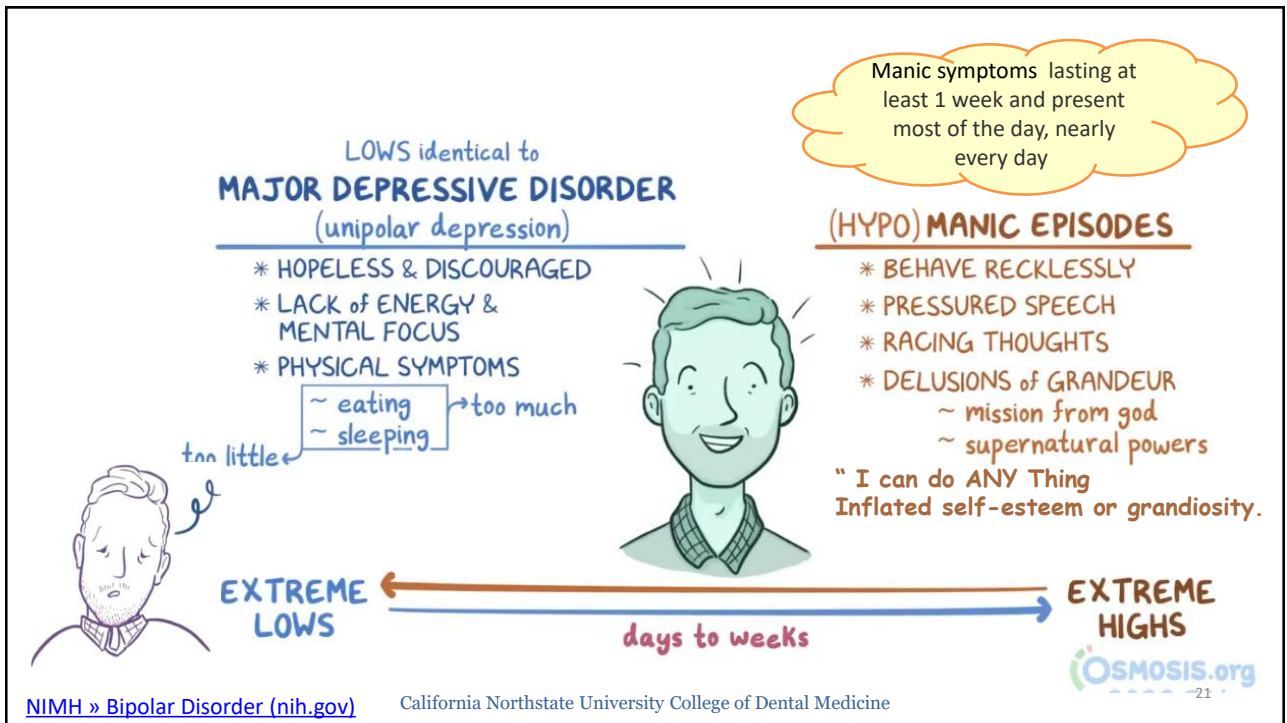
Severe mood disorder that causes the person to shift between two opposite states with respect to emotions, mood and energy levels.

According to DSM-V,

- **A manic** episode is defined as “a distinct period of abnormally and persistently elevated, expansive, or irritable mood and abnormally and persistently increased activity or energy, lasting at least one week and present most of the day, nearly every day (or any duration if hospitalization is necessary”.
- **Depressive episodes** are characterized by depressed mood, lack of interest or focus or self esteem, fatigue and appetite changes.

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20



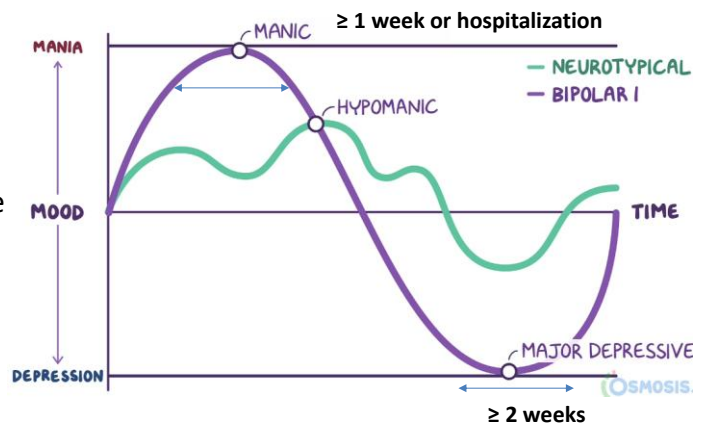
Risk factors

A combination of genetic factors, environmental factors (including drugs) and altered brain structure contribute to the development of bipolar disorder.

Types of bipolar disorder:

1- Bipolar I disorder:

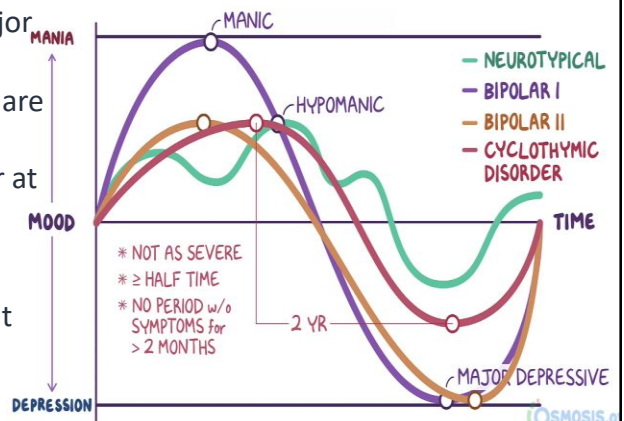
Intense manic episodes that typically lasts for a week followed by depressive episodes and those typically last for weeks. Patients of bipolar I can also experience hypomanic episodes.



Types of bipolar disorder:

2- Bipolar II disorder: cycling between major depressive episodes and hypomanic episodes without manic episodes. Hypomanic episodes are less severe than the manic episodes, do not require hospitalization and they should last for at least 4 days.

3- Cyclothymic disorder: recurrent hypomanic and mild depressive symptoms that last for at least two years. Symptoms are not intense enough or do not last long enough to qualify as manic or depressive episodes.



In general, the frequency of those high-low cycles depend on the disease severity. But typically, patients will have one or two cycles per year

Management strategies

- Pharmacological approaches

Lithium or valproate along with other anti-psychotics to control both phases of the disease.

- Non-pharmacological approaches

Non-pharmacological approaches may not be effective during manic episodes but psychotherapy like cognitive behavioral therapy can teach patient how to better handle challenging situations to prevent stressors leading to manic episodes

Anxiety disorders



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Generalized Anxiety Disorder (GAD)

- **Prevalence:** In the USA, GAD affects 6.8 million adults or 3.1% of the population. This condition is twice as frequent in females than males.
- GAD is characterized by excessive, persistent and unreasonable worry about a variety of events or activities (e.g., work or school performance). According to the severity of symptoms, categorized into mild, moderate and severe.
- GAD patients can suffer from additional mental illnesses like MDD, or phobias (social, dental, heights,,,,,).
- Substance use can be common in patients with GAD (commonly, alcohol and sedatives).

<https://adaa.org/understanding-anxiety/generalized-anxiety-disorder-gad>

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26

Onset:

Usually before age 20 years.

Diagnosis:

- Patients are diagnosed with GAD if ≥ 3 anxiety symptoms that persists for most days, for at least 6 months.
- Extreme anxiety that can cause impairment in social, occupational life.
- Anxiety cannot be controlled.
- Anxiety is not caused by substance use or other mental illness.

GAD symptoms



Behavioral symptoms: restlessness, irritability



Cognitive symptoms: inability to focus on-task or concentrate.



Physical symptoms: tiredness, sweating, palpitation, trembling, unexplained pain, headache



Other symptoms: worrying, inability to sleep.

Risk factors

- Female gender.
- Presence of stressors or negative life experiences.
- Familial history of anxiety.
- Childhood trauma and maltreatment or sexual abuse.
- Dysregulation of the locus coeruleus.
- Dysregulation of neurotransmitters (GABA, Serotonin, Noradrenaline)

Management

Pharmacological and non-pharmacological treatments.

Cognitive behavioral therapy is an effective psychotherapy option for GAD. It teaches patients coping skills to manage worries and return to functionality.

Panic disorder

- ❖ Panic disorder is a form of intense anxiety episode with a life-time prevalence of 5% in women and 2% in men. Most often, it occurs in adolescence and young adulthood.
- ❖ Recurrent panic attacks without trigger and without stressful stimuli.
- ❖ Individuals with panic disorder do not have control over panic attacks.

- Panic disorder can come with or without agoraphobia.
 - Agoraphobia is the fear of going out to places in which the individual cannot escape, or fears being trapped or fear from being in an open space all alone.
 - Fear of being trapped include being in public transportation, standing in line, movie theatres.
 - Fear of open spaces can include parking lots, stores, etc.
- Agoraphobia can be severe leading the person to become homebound.

Panic attack involves the intense feeling of threat or danger accompanied by physiological symptoms that include feelings that everything is out of control and



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Diagnosis

Patients are diagnosed with panic disorders if :

A- Patients have recurrent panic attacks within a one-month duration.

B- Attacks have abrupt onset or surge of intense fear that peaks within minutes and comes without triggers.

C- Patients should have persistent worrying or changes in behavior due to their panic attacks.

D- ≥ 4 physiological symptoms (previous slide).

E- Not an effect of substance or other disorders.

Management

Goals of treatment is to decrease the frequency of panic attacks and to reduce their intensity.

Non-pharmacological approaches

Psychotherapy like cognitive behavioral interventions and relaxation exercises.

Pharmacological approaches

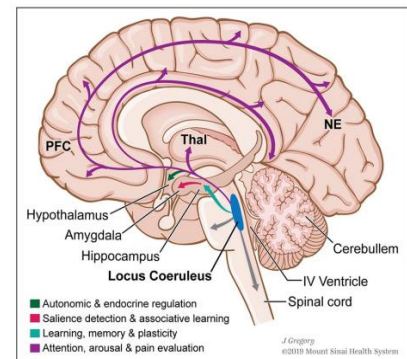
- Medications like SSRIs (antidepressants).
- Benzodiazepines can be also useful for acute management of panic attack, but they are not usually used on the long-term.

Clinical impact of Fear & Anxiety

“The Locus Coeruleus is our reptilian antenna for danger”.

Locus Coeruleus (brain stem) has norepinephrine projections throughout the CNS. Can modulate the cortical neurocognitive information and adaptive behavior.

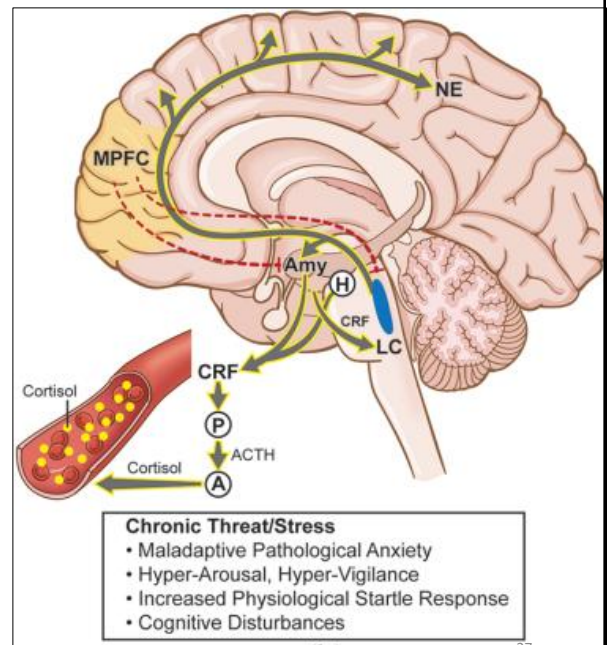
Feeling anxious and acute fear can be associated with tachycardia, increased alertness and diaphoresis. Due to the activation of SNS via the locus coeruleus.



[The role of the locus coeruleus in the generation of pathological anxiety](#) California Northstate University College of Dental Medicine

36

- Acute threat/stress increase noradrenergic discharges in the locus coeruleus. This leads to release increased NE & cortisol yielding pathological anxiety responses.
- Chronic stress leads to Locus Coeruleus hyperactivity and dysregulation yielding exaggerated fear or pathological anxiety responses.



[The role of the locus coeruleus in the generation of pathological anxiety](#)

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Impact of anxiety on pain

- In general, anxiety and stress can have negative effect on all organs.
- It enhances pain perception “Anticipation of pain augments pain perception”.
- Anxiety can present as appearing frightened or irritated.
- Anxiety can lead to “illness anxiety disorder” characterized by excessive worry that any minor body symptoms is serious or life-threatening medical condition.

Psychotic Disorders

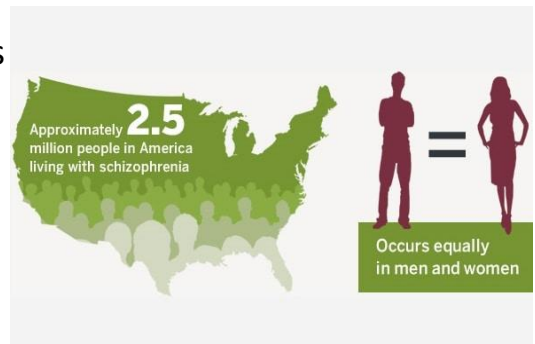
Schizophrenia

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39

Schizophrenia

- Chronic psychiatric disorder characterized by disruption in social/emotional functions accompanied by fragmented pattern of thinking.
- Age of onset is earlier in males (18 to 25 years old) than females (25-35 years).
- Schizophrenia affects approximately 2.5 million individuals in the US with equal prevalence between males and females.



Adapted from: Top 10 Schizophrenia Facts - Symptoms, Types, Causes & More- Facts.net

Symptoms

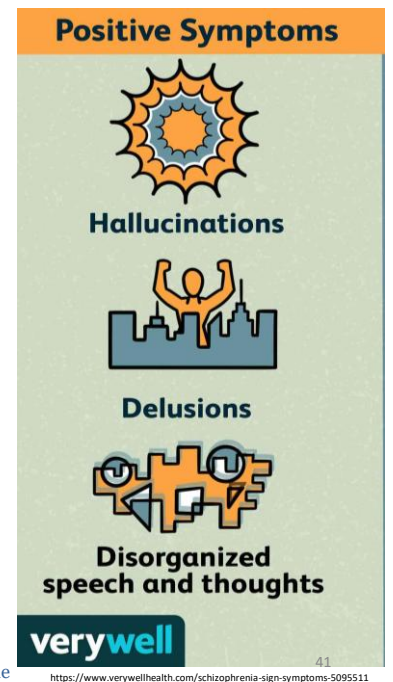
(Positive, negative and cognitive)

1- Positive symptoms

Psychotic symptoms including:

- Hallucinations (auditory hallucinations mainly in schizophrenia patients)
- Delusions (e.g., paranoia)
- Thought disorder (unusual thinking or disorganized speech).

The term positive = those symptoms won't be observed in neurotypical individuals.



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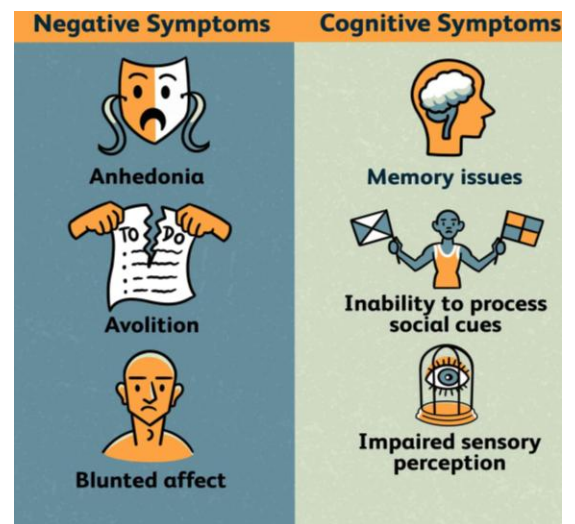
<https://www.verywellhealth.com/schizophrenia-sign-symptoms-5095511>

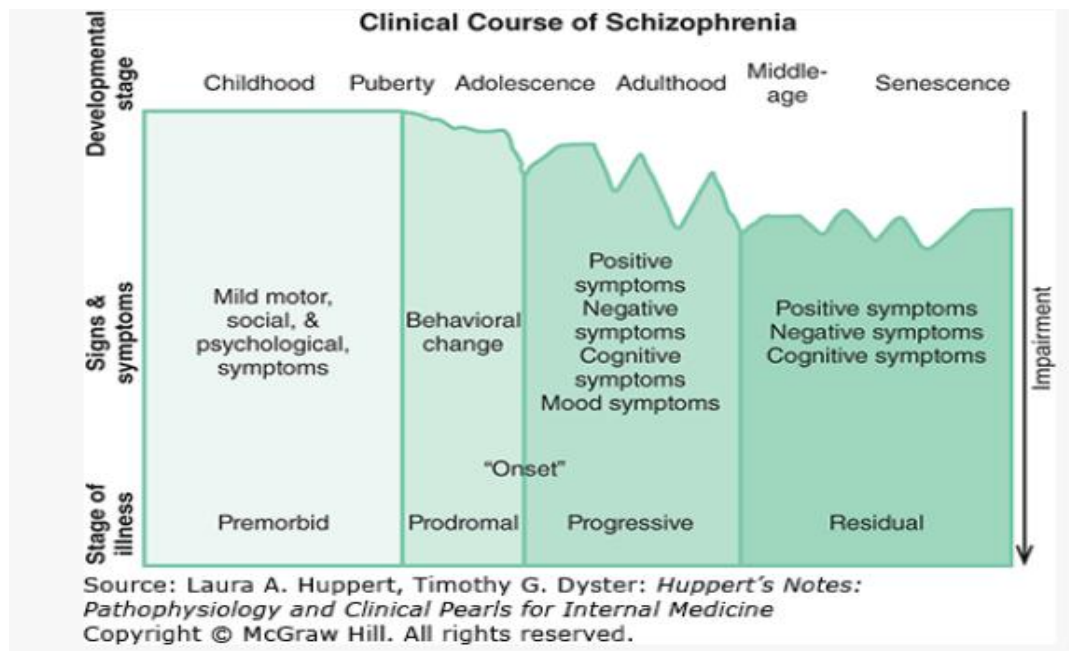
2-Negative symptoms

including social withdrawal, anhedonia, avolition which refers to loss of motivation and decreased interest in goal-directed activities, and blunted affect that refers to the reduced expression of emotions.

3- Cognitive symptoms

including problems in attention, concentration, and memory. These are subtle or hard to notice.





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43

Clinical Phases of disease

- 1) Prodromal phase: behavioral change like social withdrawal, isolation
- 2) Active phase characterized by active positive, negative and cognitive symptoms.
- 3) Residual phase is the period that persists between active episodes.

Diagnosis

At least one positive symptom (delusions, hallucinations, or disorganized speech) + an additional negative or positive symptom. The two symptoms must be continuously observed for 1-month period and the patient must exhibit persistent disturbance for at least 6 months.

Risk factors

1- Abnormalities in the brain structure :

Enlargement of the cerebral ventricles and thinner cortex were associated with increased risk of schizophrenia.

2- Increased dopamine level.

3- Environmental factors:

Birthing complications, severe maternal malnutrition, childhood trauma, social isolation, chronic cannabis use.

4- Genetic factors:

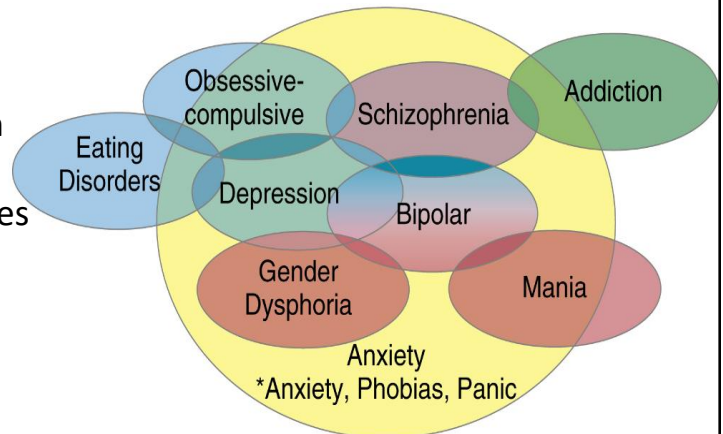
Several gene mutations were identified, like: *COMT* (catechol-O-methyltransferase) gene mutations increase the schizophrenia risk by 30 times via increasing the dopamine level.

Pathophysiology: Dopamine hypothesis

1- Neurochemical abnormality:

Imbalance of dopamine, serotonin and glutamate.

The dopamine hypothesis attributes schizophrenia symptoms to the hyperactivity of central dopamine pathways.



2- Neuroanatomical changes: Reduction in grey matter volume was observed in schizophrenia patients.

3- Neurodevelopmental changes:

- developmental grey and white matter abnormalities were observed, and they can contribute to the motor and cognitive impairments
- Few studies observed that palate width and depth was significantly higher in schizophrenia patients.

[Palate size and shape in schizophrenia – ScienceDirect](#)

[Morphology of the palate, palatal rugae pattern, and dental arch form in patients with schizophrenia - PubMed \(nih.gov\)](#)

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47

Management strategies

1- Non-pharmacological approach:

- Family education and support.
- Psychosocial treatment like cognitive behavioral therapy and behavioral skills training can help with sparking the patient interest and enhance their coping skills.

2- Pharmacological approach “Antipsychotic medications”.

- First generation antipsychotics
- Second generation antipsychotics.

In a field where you can be anything, choose to be kind

And access to mental health care remains out of reach for millions of people in the U.S. who need it.

In 2022 and 2023:

77.09%

of all **adults** with a **substance use disorder** did **not receive treatment**.

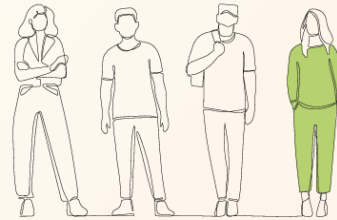
9.2%

of **adults** with **any mental illness (AMI)** were **uninsured**, totaling over 5 million people.

28.6%

of **adolescents** (ages 12-17) **did not have a preventive health visit** in the past year.

That totaled over 7 million young people in the U.S.



1 in 4 (25%) adults with AMI reported an **unmet need for mental health treatment** in the past year.

<https://mhanational.org/the-state-of-mental-health-in-america/>

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49



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50

People First Language

Put the person first before the disability. It aims to acknowledge the person, rather than the illness or disability.

Say this

Patient with Schizophrenia
Child with xxxx disability
Patient suffering from depression
Person with substance use disorder
(or addiction or opiate use).

Rather than

Schizophrenic patient
Disabled child
John Doe, 25, depressed...
Drug addict

Resources

